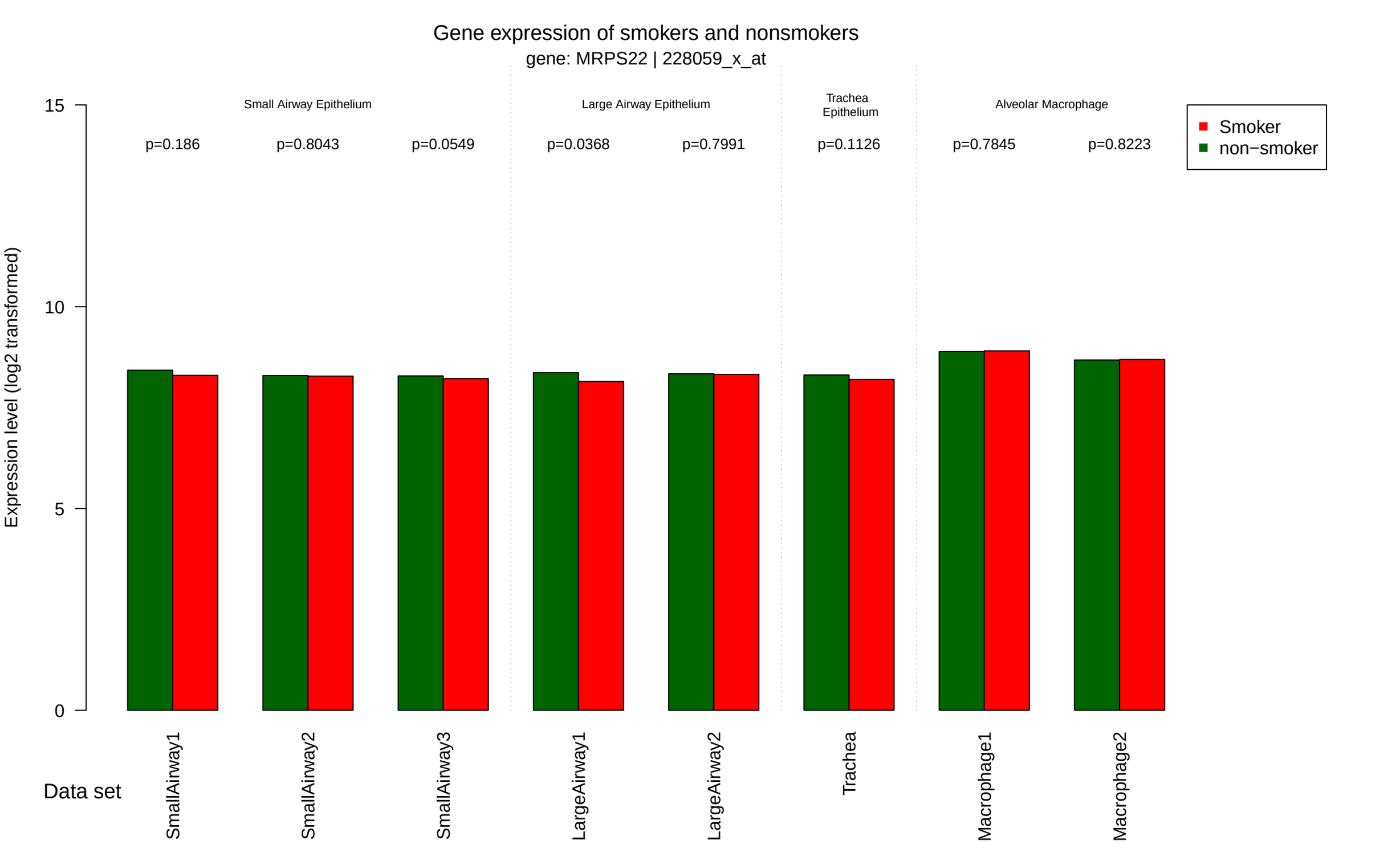
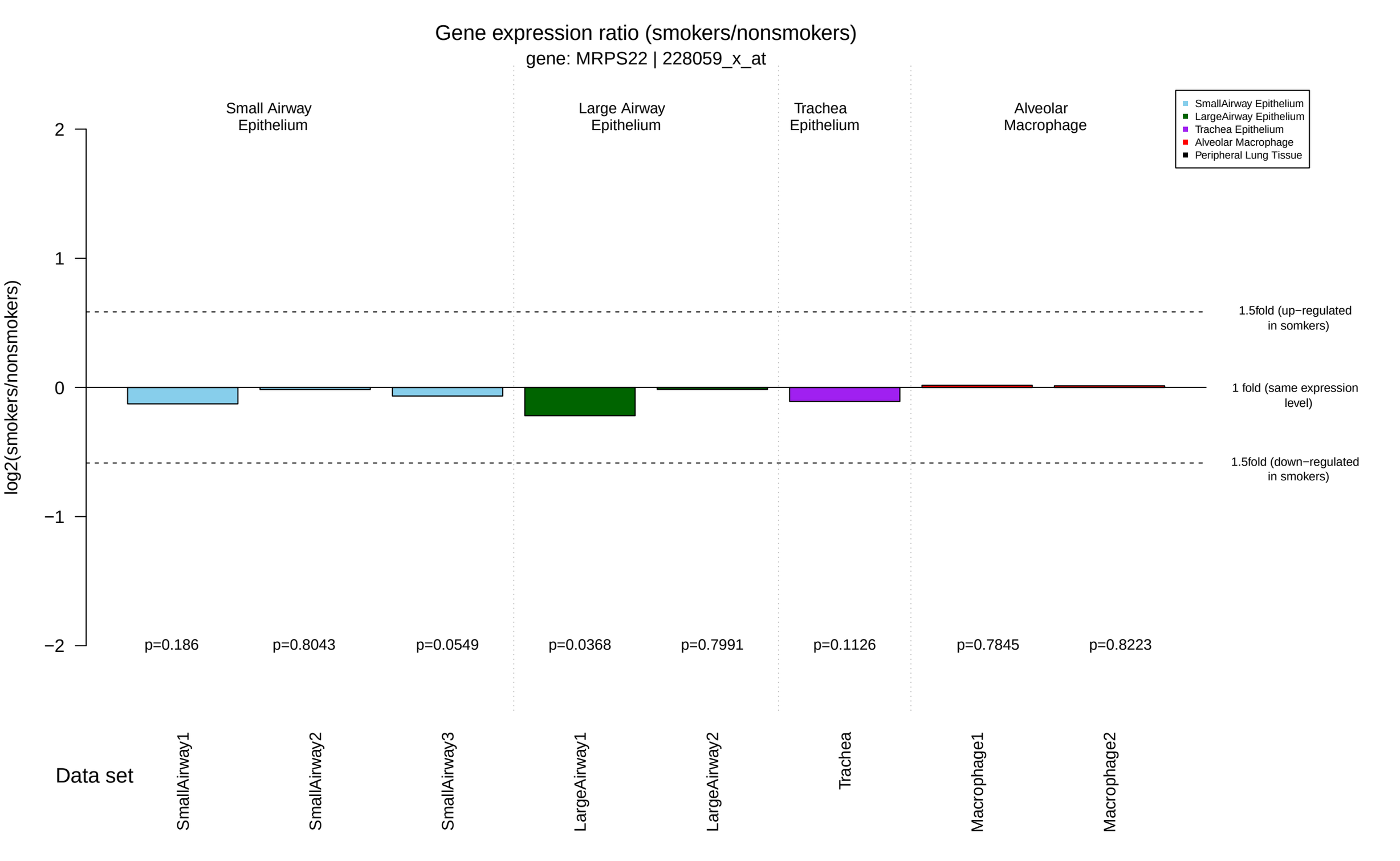
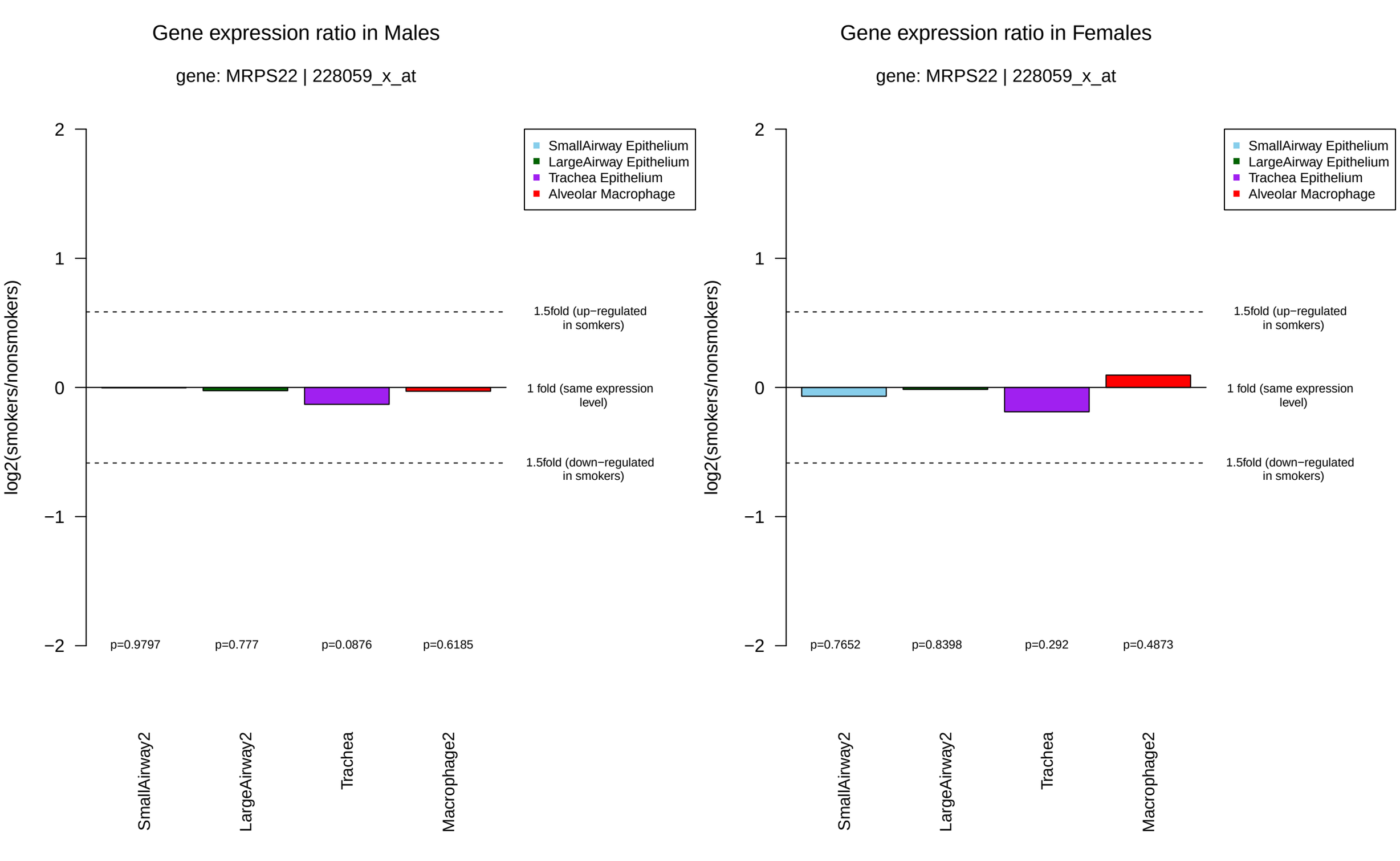
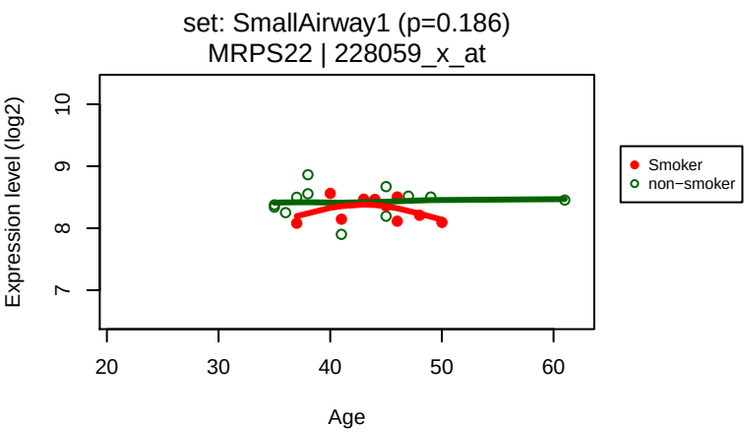
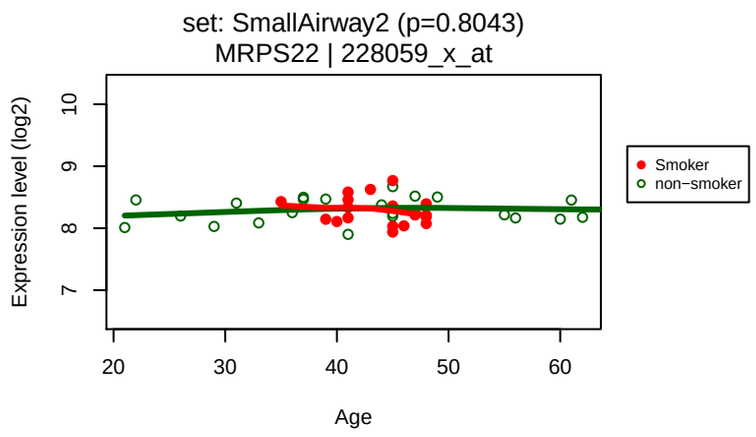
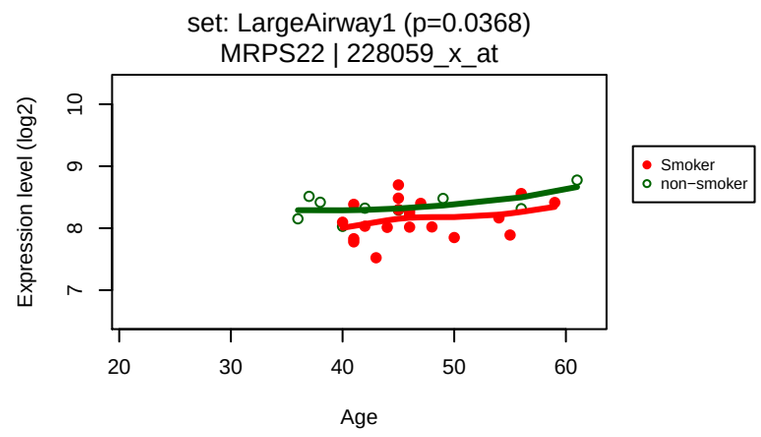

Gene expression of smokers and nonsmokers
gene: MRPS22 | 228059_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.186
p=0.8043
p=0.0549
p=0.0368
p=0.7991
p=0.1126
p=0.7845
p=0.8223
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

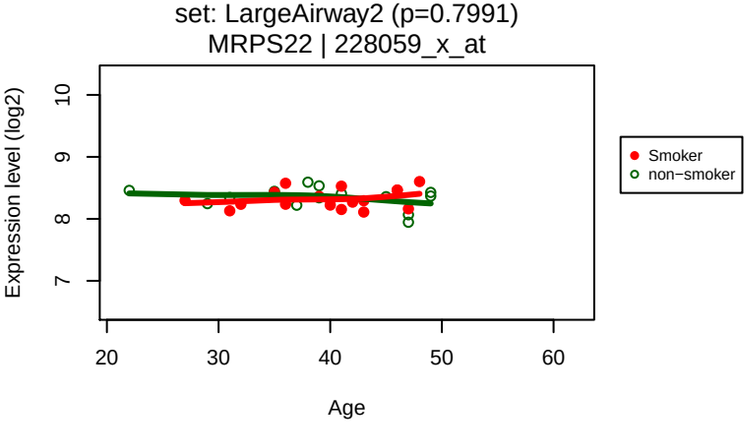

Gene expression ratio (smokers/nonsmokers)
gene: MRPS22 | 228059_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.186
p=0.8043
p=0.0549
p=0.0368
p=0.7991
p=0.1126
p=0.7845
p=0.8223
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

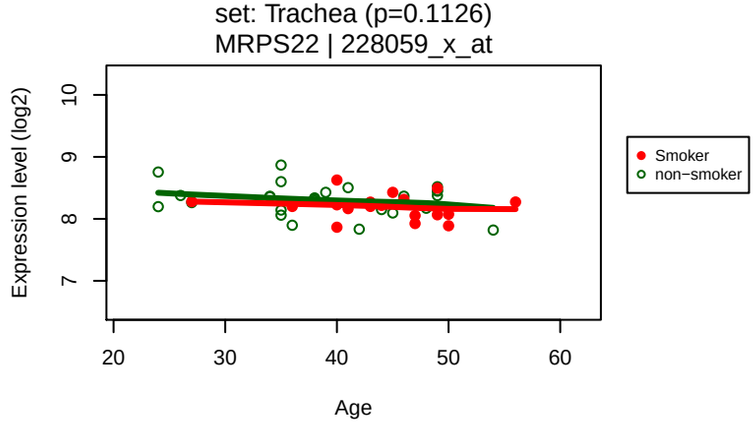

Gene expression ratio in Males
gene: MRPS22 | 228059_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9797
p=0.777
p=0.0876
p=0.6185
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPS22 | 228059_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7652
p=0.8398
p=0.292
p=0.4873
SmallAirway2
LargeAirway2
Trachea
Macrophage2

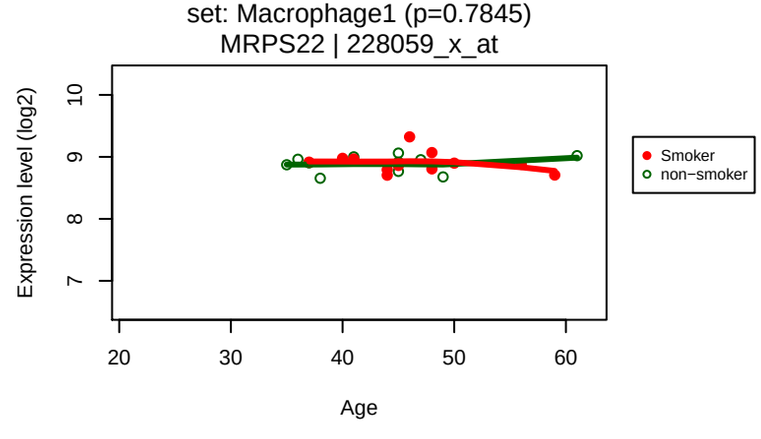

set: SmallAirway1 (p=0.186)
MRPS22 | 228059_x_at
Expression level (log2)
Age
Smoker
non−smoker

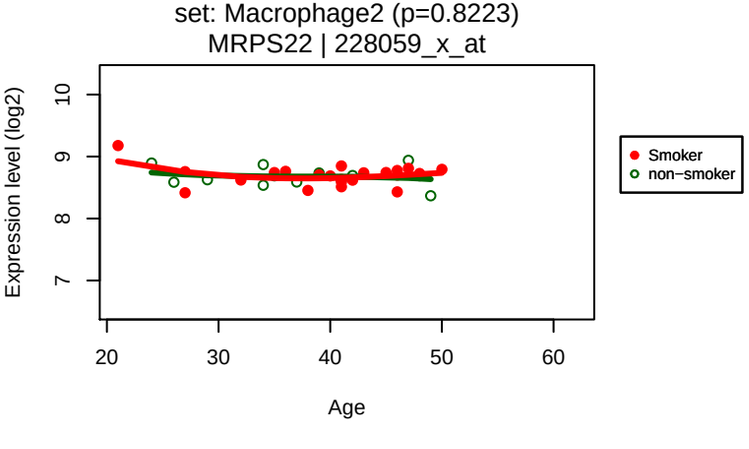

set: SmallAirway2 (p=0.8043)
MRPS22 | 228059_x_at
Expression level (log2)
Age
Smoker
non−smoker

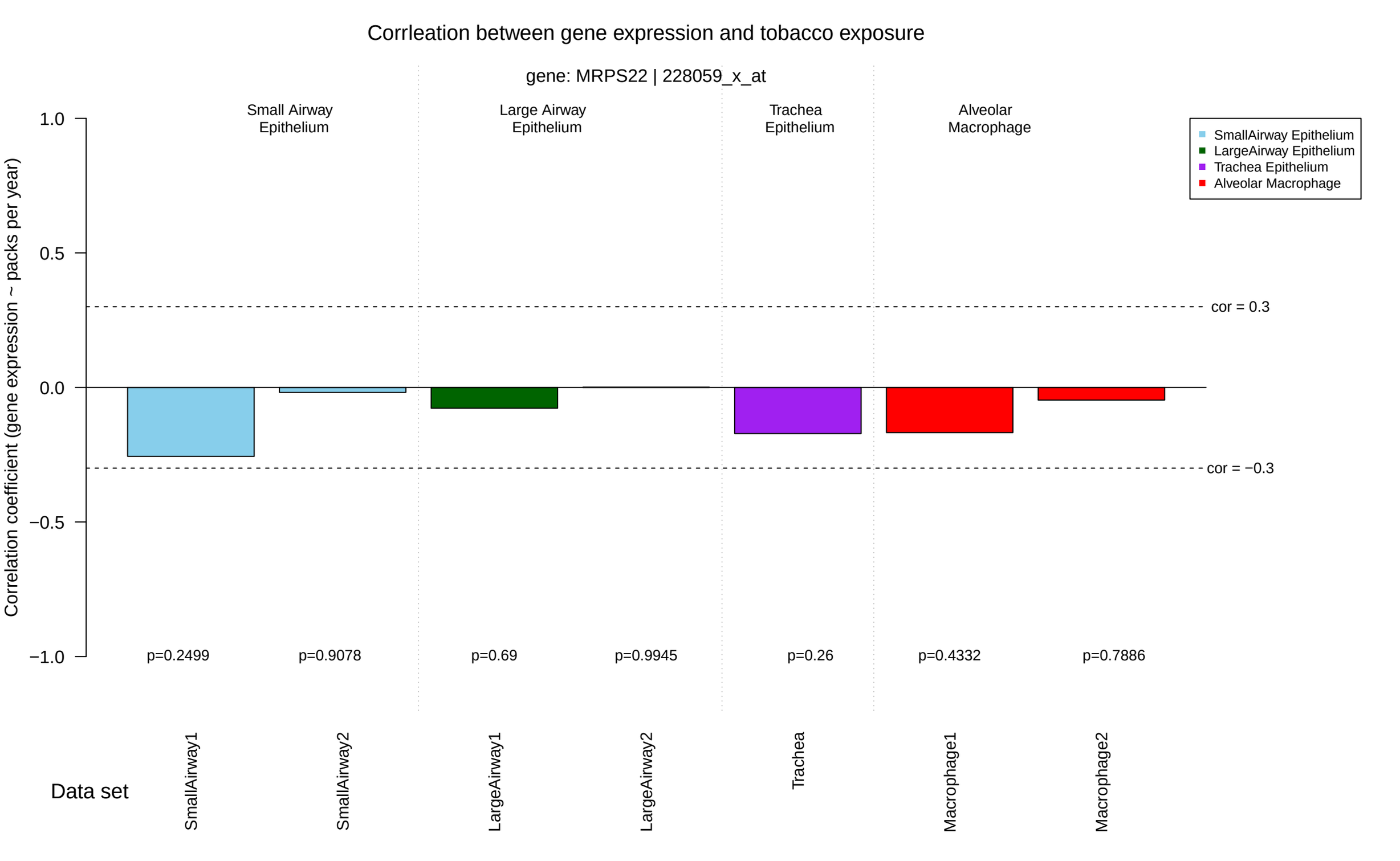

set: LargeAirway1 (p=0.0368)
MRPS22 | 228059_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7991)
MRPS22 | 228059_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1126)
MRPS22 | 228059_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7845)
MRPS22 | 228059_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8223)
MRPS22 | 228059_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPS22 | 228059_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2499
p=0.9078
p=0.69
p=0.9945
p=0.26
p=0.4332
p=0.7886
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2